

COSMOS-P10

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Boot identity of an AI runtime root by sentinel content where existence of a directory is not identity**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P10**Kind:** System (narrow)**Status:** FILE (narrow)**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to identifying a runtime root of an AI operating system by sentinel file content rather than by path existence, parent-walking, or a fallback ladder of drives.

BACKGROUND OF THE INVENTION

[0003] Empty-dir scar: a path that exists is treated as the install. Fallback ladders silently pick the wrong tree. Two writers on two roots. That is a deletion class when the wrong tree is overwritten.

BRIEF SUMMARY OF THE INVENTION

[0004] At boot, resolve one root by reading a sentinel file content, not by path existence. Match declared system and tree_id; refuse READY on mismatch or missing install record. Do not walk parents, do not try a list of drives, do not treat an empty directory as identity.

DEFINITIONS

[0005] As used herein, "Sentinel" means A file whose content (not its path) identifies the runtime root.

[0006] As used herein, "READY" means The service state that may serve. Forbidden without sentinel match.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows a boot resolver reading sentinel file content (.cosmos-root.json system and tree_id). An empty directory that exists is marked NOT IDENTITY. Parent-walk and drive-literal fallbacks are marked FORBIDDEN.

[0008] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0009] At boot, resolve one root by reading a sentinel file content, not by path existence.

[0010] Preferred sentinel: .cosmos-root.json declaring system=COSMOS and a tree_id. Preferred live tree_id: KMesh-COSMOS-live.

[0011] Match declared system and tree_id against the install record. Refuse READY on mismatch or missing install record.

[0012] Do not walk parents. Do not try a list of drives. Do not treat an empty directory as identity.

[0013] No import-time side effects. The resolver is instantiated at boot composition.

[0014] Two roots must not be conflated: the repo tree (tracked code) and the runtime root (live/). Existence of a folder is not identity of either.

[0015] Every path resolves through a declared role under that root. Nothing assembles a path by hand from a drive letter.

BEST MODE

[0016] cosmos_paths resolver. Service cannot go READY without sentinel-verified root. AD-5. Encoded cosmos/cosmos_paths.py.

FURTHER EMBODIMENTS

[0017] A unit test plants an empty directory at a tempting path and asserts the resolver refuses it.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0018] Resolver rules in public architecture 2026-08-23.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0019] OCI digest versus tag; Nix require-sigs / sandbox-fallback (fallback is the anti-pattern); git objects; chroot (path, not content); systemd-nspawn --root-hash; SLSA/in-toto/Cosign; HSTS fail-closed. No close patent found on JSON sentinel plus refuse empty-dir plus no parent-walk as AI-OS boot identity. Unknown as blocking. Narrow.

WHAT THIS DISCLOSURE IS NOT

[0020] Not chroot. Not Docker tags. Not a new hash algorithm.

STATEMENT OF INVENTION (NOT CLAIMS)

[0021] Boot identity of an AI runtime root by sentinel content, where existence of a directory is not identity and fallback ladders are forbidden.

[0022] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0023] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.